Diya Patel

Thermal Systems Engineer

RESUME_1892

Location: Bhopal, Madhya Pradesh, India | Email: diya.patel.resume_1892@example.com | Phone: +91-00000-00000 (placeholder)
Total Experience: 3 years

PROFESSIONAL SUMMARY

Thermal Systems Engineer with 3 years of experience in thermal engineering, using Thermodynamics, Heat Transfer, ANSYS Fluent, CFD. Experienced in thermal design, energy efficiency and measurable delivery outcomes.

TECHNICAL SKILLS

Skills: HVAC, MATLAB, Experimental Testing, Thermodynamics, Heat Exchangers, Data Acquisition, Heat Transfer, CAD, ANSYS Fluent, CFD, Energy Audit

Methods: Thermal design, Energy efficiency, CFD validation

PROFESSIONAL EXPERIENCE

Thermal Systems Engineer | ZenithTrail Technologies | Jul 2023 - Present

- Reduced simulated hotspot temperature by 34% through airflow and interface changes
- Reduced modeled process heat loss by 34% after insulation improvements
- Executed 234 controlled thermal test cases with uncertainty estimates
- Coordinated engineering design reviews, acceptance criteria and test evidence with cross-functional stakeholders.

PROJECTS

Electronics Cooling Study | Experimental Testing, MATLAB, HVAC, ANSYS Fluent

- Evaluated heat sink geometry and forced convection response.
- Documented acceptance criteria and evaluated the design using energy efficiency and cfd validation.

Industrial Oven Energy Audit | Experimental Testing, Data Acquisition, Heat Transfer, Energy Audit

- Estimated losses, insulation upgrades and process energy balance.
- Documented acceptance criteria and evaluated the design using energy efficiency and cfd validation.

EDUCATION

B.Tech in Mechanical Engineering, Crescent Valley Technical University, 2023
Relevant Methods: Thermal design, Energy efficiency, CFD validation

CERTIFICATIONS

CFD for Thermal Design (2025)